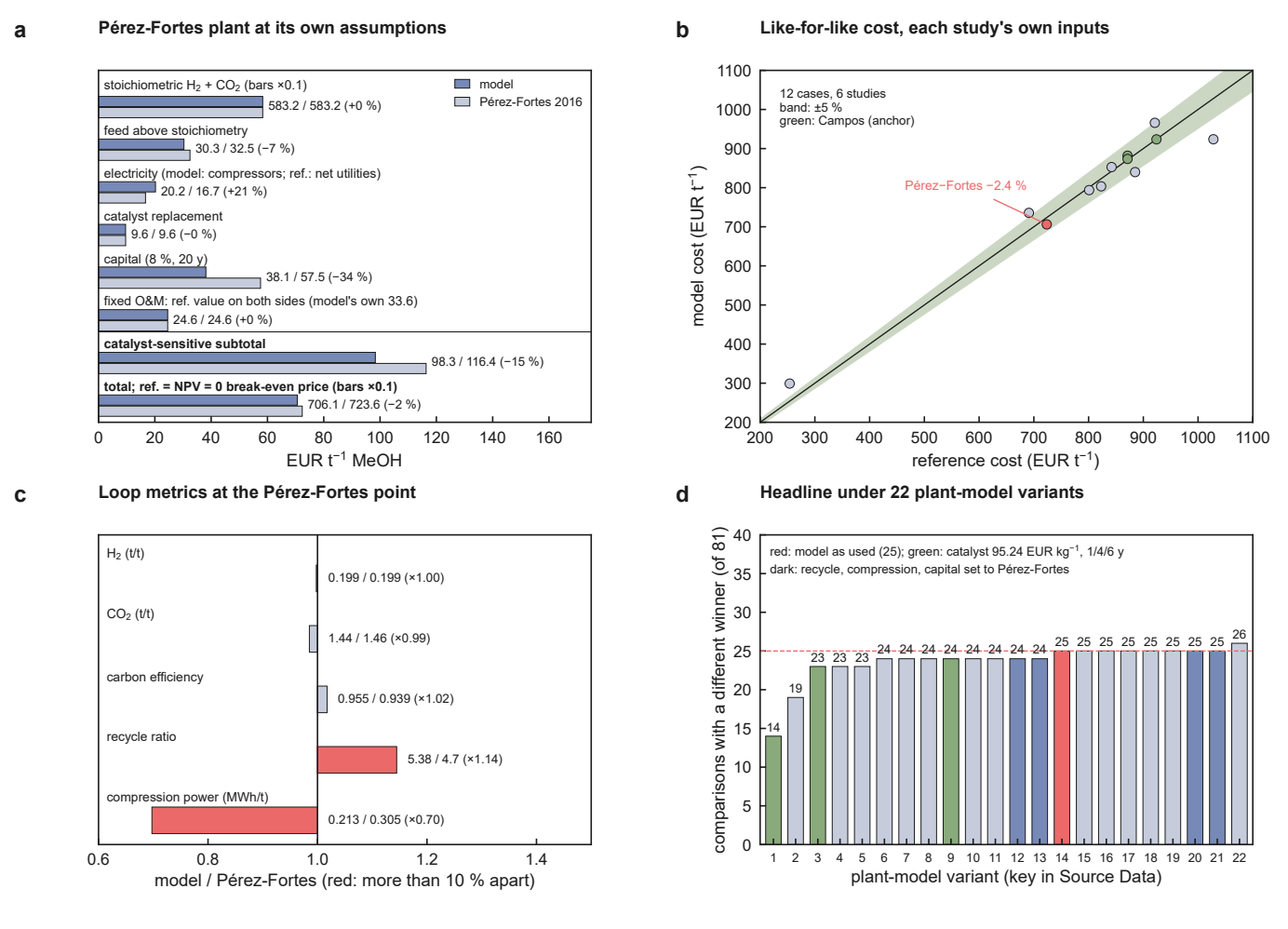

a
Pérez-Fortes plant at its own assumptions
stoichiometric H2 + CO2 (bars ×0.1)
583.2 / 583.2 (+0 %)
model
Pérez-Fortes 2016
feed above stoichiometry
30.3 / 32.5 (−7 %)
electricity (model: compressors; ref.: net utilities)
20.2 / 16.7 (+21 %)
catalyst replacement
9.6 / 9.6 (−0 %)
capital (8 %, 20 y)
38.1 / 57.5 (−34 %)
fixed O&M: ref. value on both sides (model's own 33.6)
24.6 / 24.6 (+0 %)
catalyst-sensitive subtotal
98.3 / 116.4 (−15 %)
total; ref. = NPV = 0 break-even price (bars ×0.1)
706.1 / 723.6 (−2 %)
0 20 40 60 80 100 120 140 160
EUR t−1 MeOH
b
Like-for-like cost, each study's own inputs
12 cases, 6 studies
band: ±5 %
green: Campos (anchor)
Pérez−Fortes −2.4 %
model cost (EUR t−1)
reference cost (EUR t−1)
200 300 400 500 600 700 800 900 1000 1100
c
Loop metrics at the Pérez-Fortes point
H2 (t/t)
0.199 / 0.199 (×1.00)
CO2 (t/t)
1.44 / 1.46 (×0.99)
carbon efficiency
0.955 / 0.939 (×1.02)
recycle ratio
5.38 / 4.7 (×1.14)
compression power (MWh/t)
0.213 / 0.305 (×0.70)
0.6 0.8 1.0 1.2 1.4
model / Pérez-Fortes (red: more than 10 % apart)
d
Headline under 22 plant-model variants
red: model as used (25); green: catalyst 95.24 EUR kg−1, 1/4/6 y
dark: recycle, compression, capital set to Pérez-Fortes
comparisons with a different winner (of 81)
14 19 23 23 23 24 24 24 24 24 24 24 25 25 25 25 25 25 25 26
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22
plant-model variant (key in Source Data)